

3D Object Representations

Outline

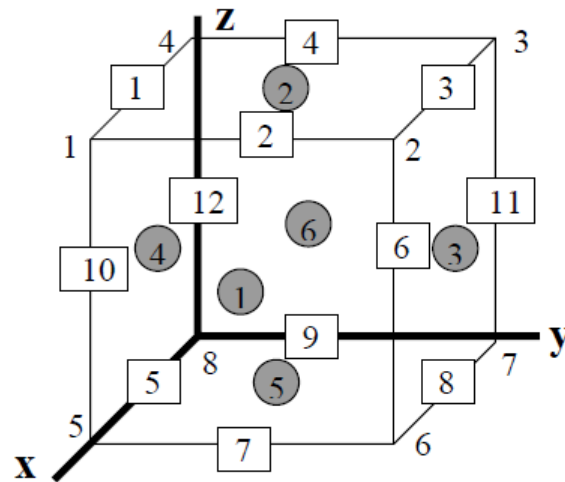
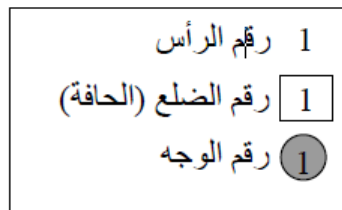
- How objects are modelled in 3D
- Boundary representation
- Solid Geometry
- Polyhedra
- Quadric surfaces
- Sweep representations
- Constructive Solid Geometry (CSG)
- Quadtree

Boundary representation

Boundary representation (B-Rep)

- Approximate 3d object to a set of surfaces:
 - $AX + BY + CZ + D = 0$
- Three building blocks
 - Vertices
 - Edges
 - Faces

Boundary representation (B-Rep)



Best representation

- Wrong representation
 - Represent each face by coordinates of vertices
- Correct representation
 - Vertex numbering
 - Represent edges and faces by indices of vertices

$nV = 8$

1	10, 0, 10
2	10, 10, 10

$nE = 12$

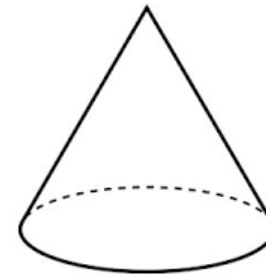
1	1	4
2	1	2

$nF = 6$

1	2	6	7	10
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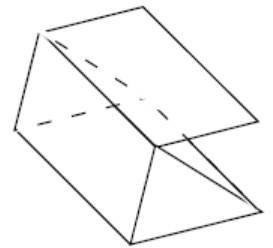
Example

- How much storage to store this cone approximated to 256 edge pyramid
- • Integer 4 byte, fractional 8 byte
- • $nV=257$, $nF=257$ $nE=256*2=512$
-
- $257 * 3 * 8\text{bytes} + 512 * 2 * 4\text{bytes} + 256 * 3 * 4\text{bytes} + 256 * 1 * 4\text{bytes}$
 $= 14360 \text{ byte}$
- Integer 4 byte, fractional 8 byte

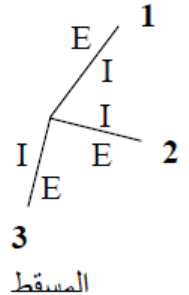


Solids

- Any realistic object
- Lines and planes are not solids (they are abstract and does not exist in reality)
- Solids has interior and exterior
- Every face of a solid object MUST separate interior and exterior
- Every edge MUST be shared by two and only two faces.



الشكل غير المصمت



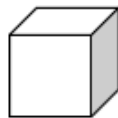
Euler rule in solids

$$V + F - E = 2$$

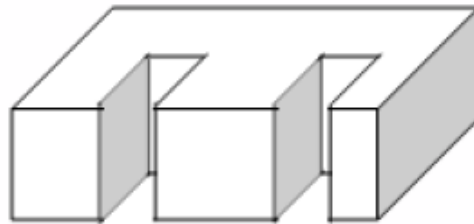
V عدد الرؤوس

F عدد الوجوه

E عدد الحواف



V	+	F	-	E	=	2
8		6		12		2



24		14		36		2
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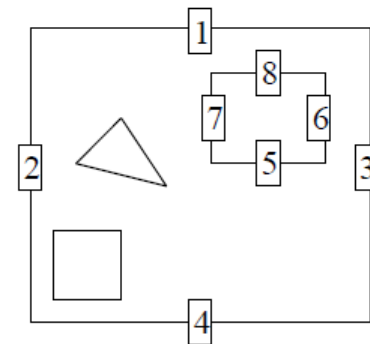
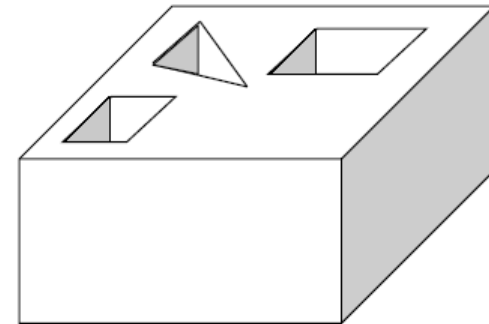
Solids

- أجسام فيزيائية أو قابلة للتحقيق فيزيائيا لها داخل (مادة)، وخارج (لا مادة).
- النقطة ليست Solid؛ تعريفها جسم ليس له أبعاد.
- المستقيم ليس solid؛ ليس له ثخانة ولا حجم.
- السطح ليس solid؛ لأنه جسم مفاهيمي ليس له أبعاد ولا حجم ولا ثخانة ويمكن تعميم كل ما هو one face لا يعتبر جسم مصمت.
- الورقة هي solid؛ لأنها تحتوي على أكثر من بعد ولها سماكة.

Representing holes

- $nV = 8 + 8 + 8 + 6 = 30$
- $nE = 12 + 12 + 12 + 9 = 45$
- $nF = 6 + 5 + 5 + 4 = 20$
- 3 holes. $T=3$

$$V + F - E - T = 2$$



Euler rule in solids

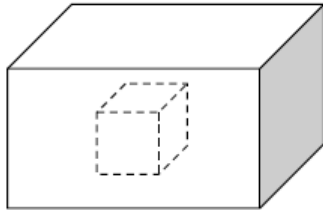
$$V + F - E - T + 2H - 2C = 2$$

C: Cave

H: Path

T: Hole

مكعب يحتوي كهف



V	+	F	-	E	-	T	+	2H	-	2C	=	2
16		12		24		0		2×0		2×1		2

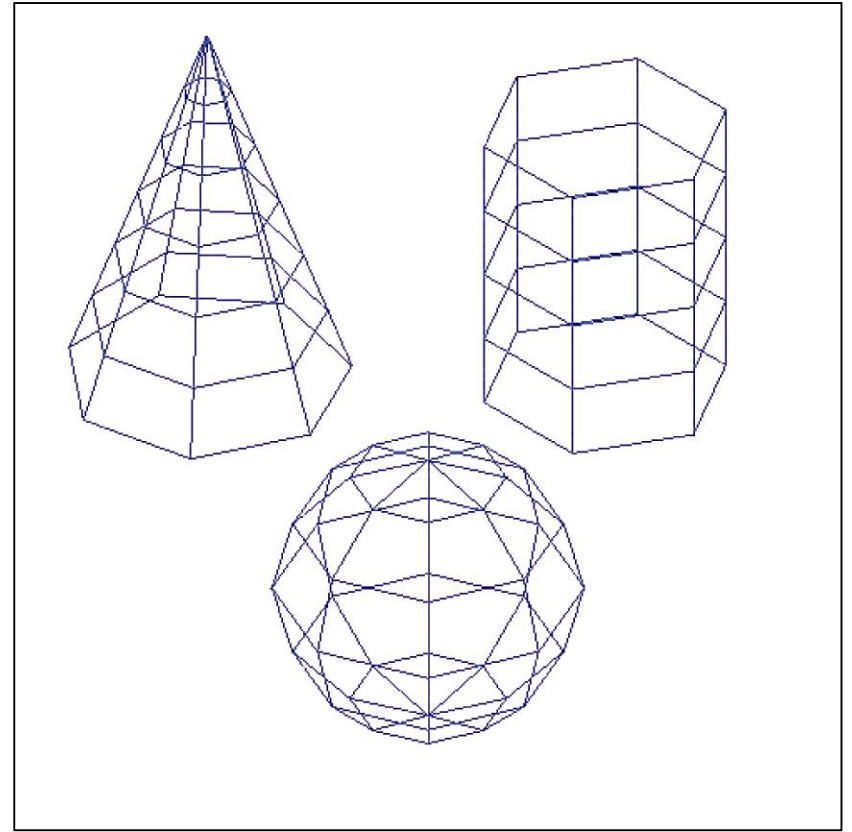
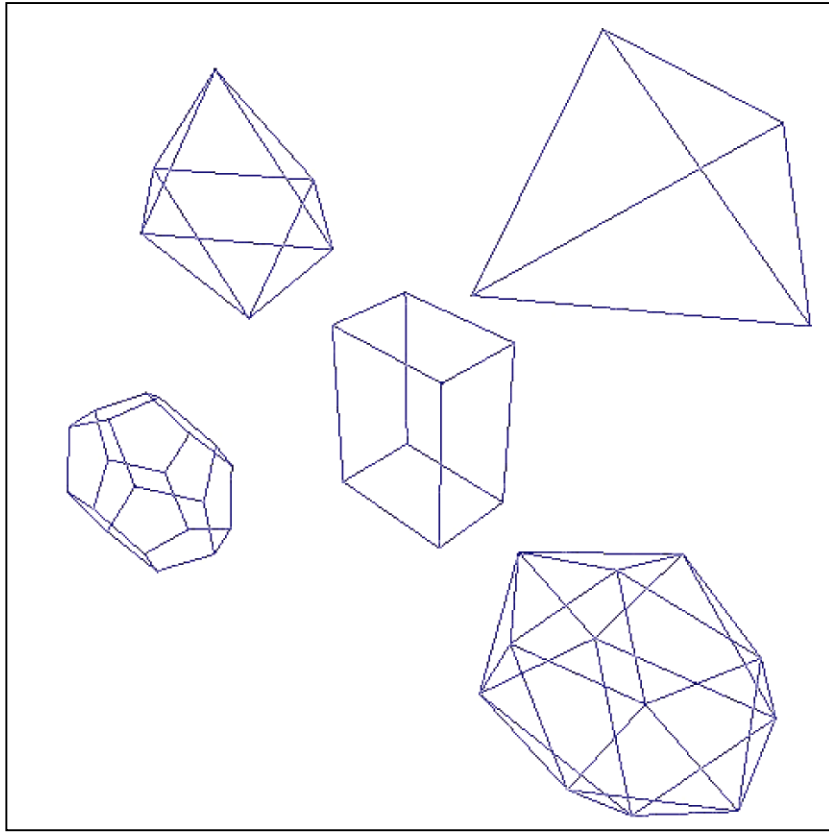
Outline

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- Polyhedra
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- Quadtree

Polyhedra

- Objects are simply a set of surface polygons that enclose an object interior
- Simplest and fastest way to render objects
- Often referred to as standard graphics objects
- In many cases packages allow us to define objects as curved surfaces etc but actually convert these to polygon meshes for display
- To define polyhedra we simply define the vertices of the polygons required

Polyhedra (contd.)



Quadric Surfaces

- A frequently used class of objects are quadric surfaces
- These are 3D surfaces described using quadratic equations
- Quadric surfaces include:
 - Spheres
 - Ellipsoids
 - Torus
 - Paraboloids
 - Hyperboloids

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Quadric Surfaces - Spheres

- A spherical surface with radius r centred on the origin is defined as the set of points (x, y, z) that satisfy the equation

$$x^2 + y^2 + z^2 = r^2$$

- This can also be done in parametric form using latitude and longitude angles

$$x = r \cos \phi \cos \theta$$

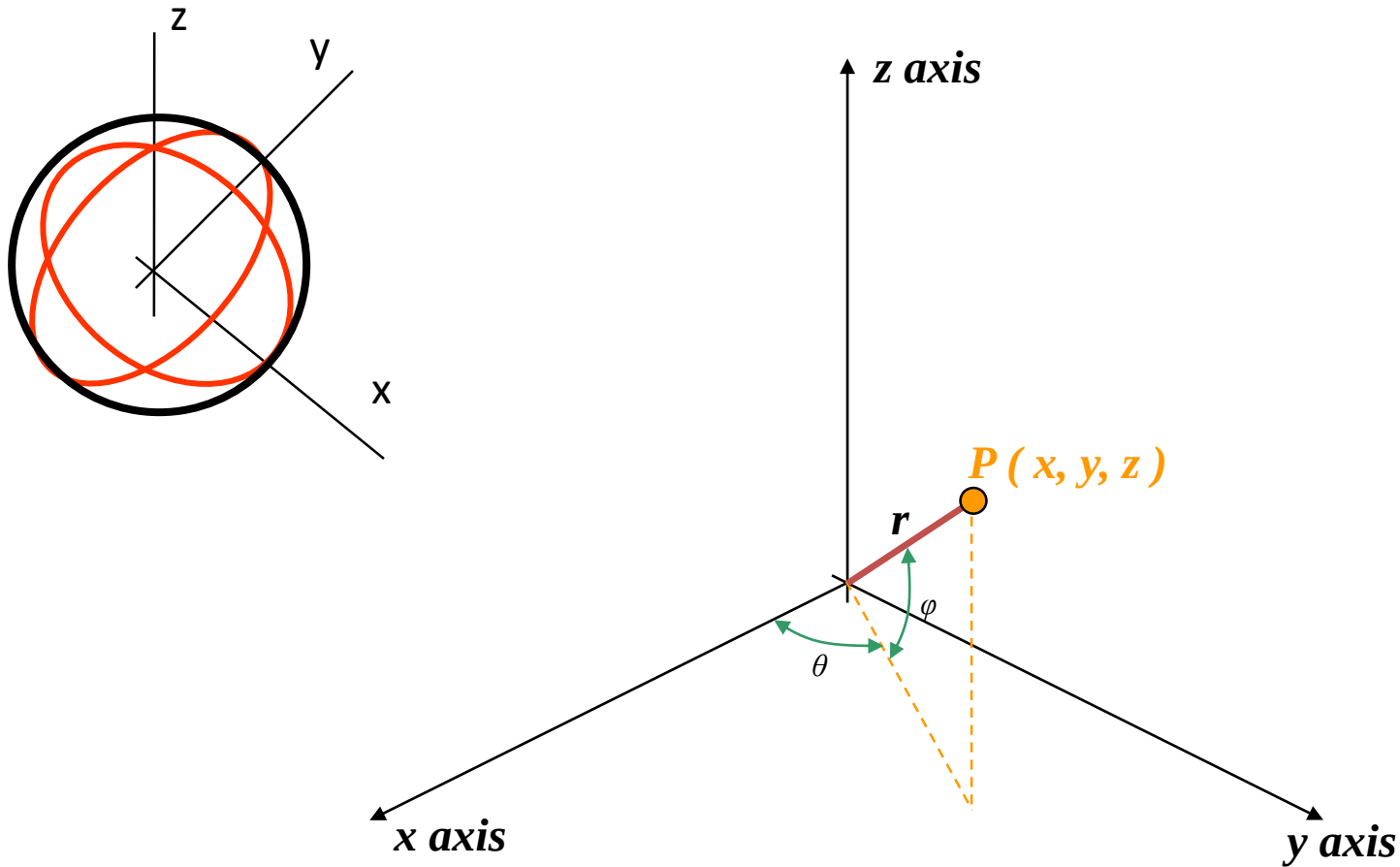
$$y = r \cos \phi \sin \theta$$

$$z = r \sin \phi$$

$$-\pi/2 \leq \phi \leq \pi/2$$

$$-\pi \leq \theta \leq \pi$$

Quadric Surfaces – Spheres (Contd.)

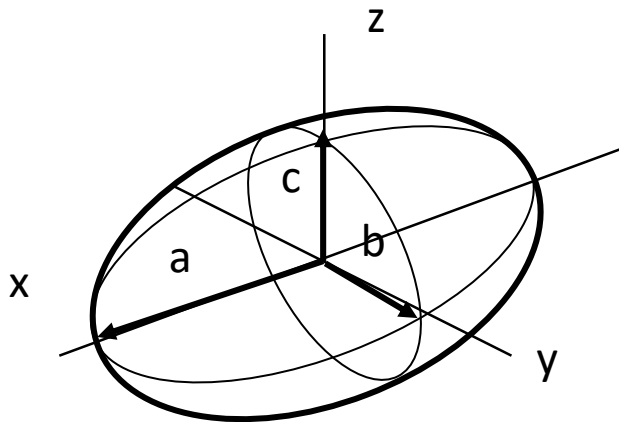


Quadratic surfaces 2

Ellipsoid: scaled sphere (mentos, cigar)

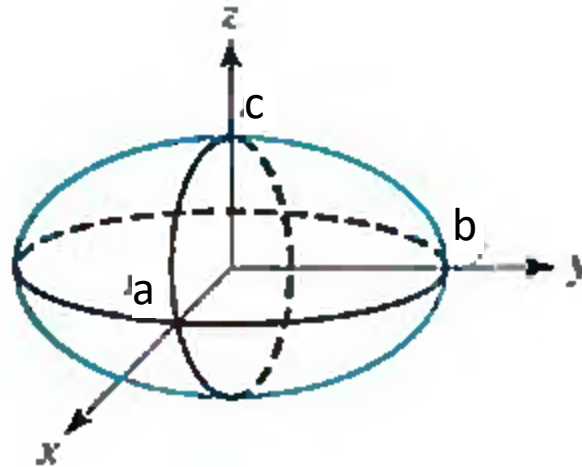
sphere: $f(x, y, z) = x^2 + y^2 + z^2 - r^2 = 0$

ellipsoid: $f(x, y, z) = \left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 + \left(\frac{z}{c}\right)^2 - 1 = 0$



Ellipsoid

- Ellipsoidal surface can be described as an extension of a spherical surfaces.



$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

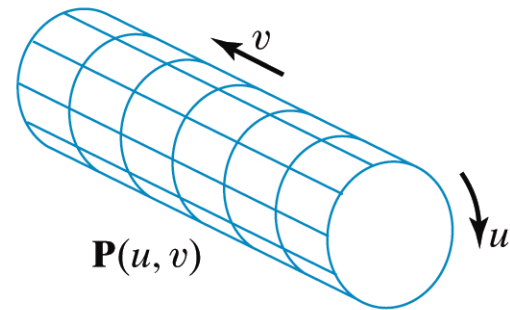
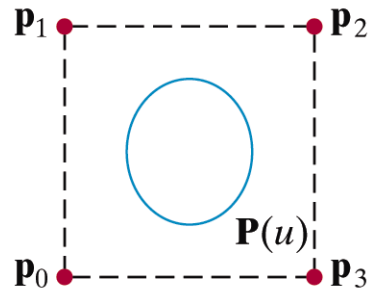
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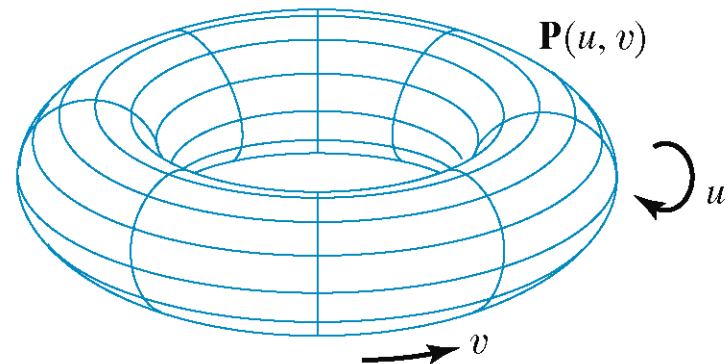
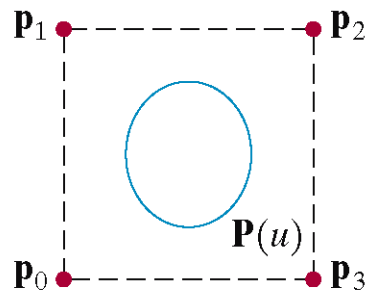
Sweep Representations

- Sweep representations are useful for constructing 3 dimensional objects that possess translational, rotational or other symmetries
- Objects are specified as a 2 dimensional shape and a sweep that moves that shape through a region of space

Sweep Representations - Examples



Axis of
Rotation



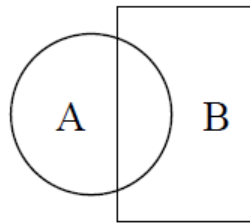
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Constructive Solid Geometry Methods

- Constructive Solid Geometry (CSG) performs solid modeling by generating a new object from two three dimensional objects using a set operation
- Valid set operations include
 - Union
 - Intersection
 - Difference

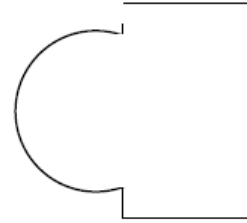
Boolean operators in CSG



$A - B$



$A \cap B$



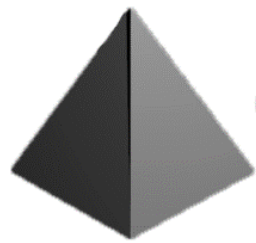
$A \cup B$



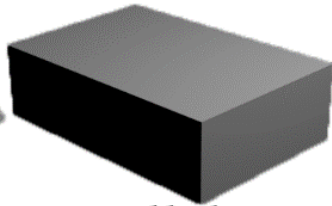
$B - A$

Difference intersection union

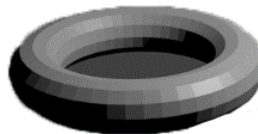
Some Primitives



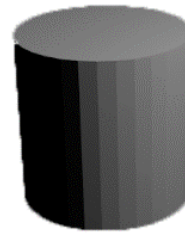
Pyramid



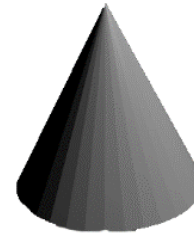
Parallelogram



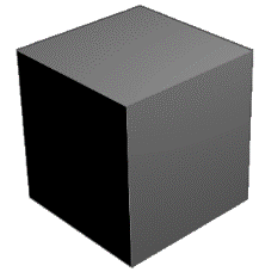
Tyre



Cylender



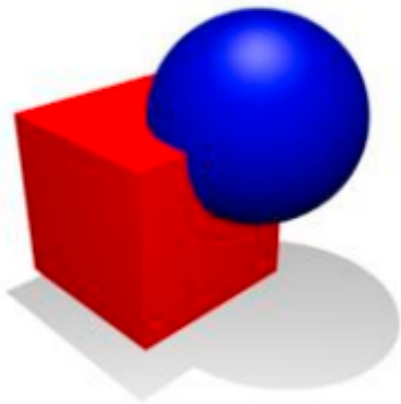
Con



Cube

Boolean operations in CSG

بفرض لدينا كرة ومكعب، يمكن تطبيق المعاملات البوليانية بينها، بالشكل التالي:



الاجتماع - $\cup$ - Union



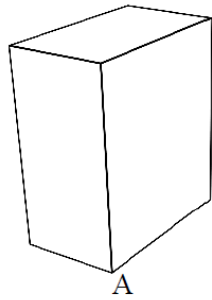
الفرق - $/$ - Difference



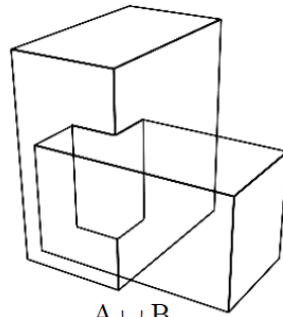
التقاطع - $\cap$ - Intersection

بتوظيف المعاملات البوليانية، يمكن الوصول إلى أشكال أكثر تعقيداً، انطلاقاً من الأشكال البسيطة.

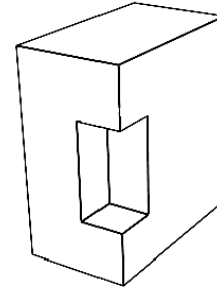
Boolean operators in CSG



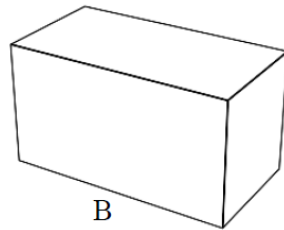
A



$A \cup B$



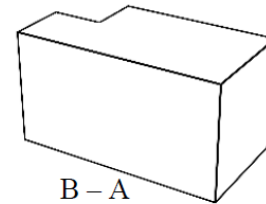
$A - B$



B

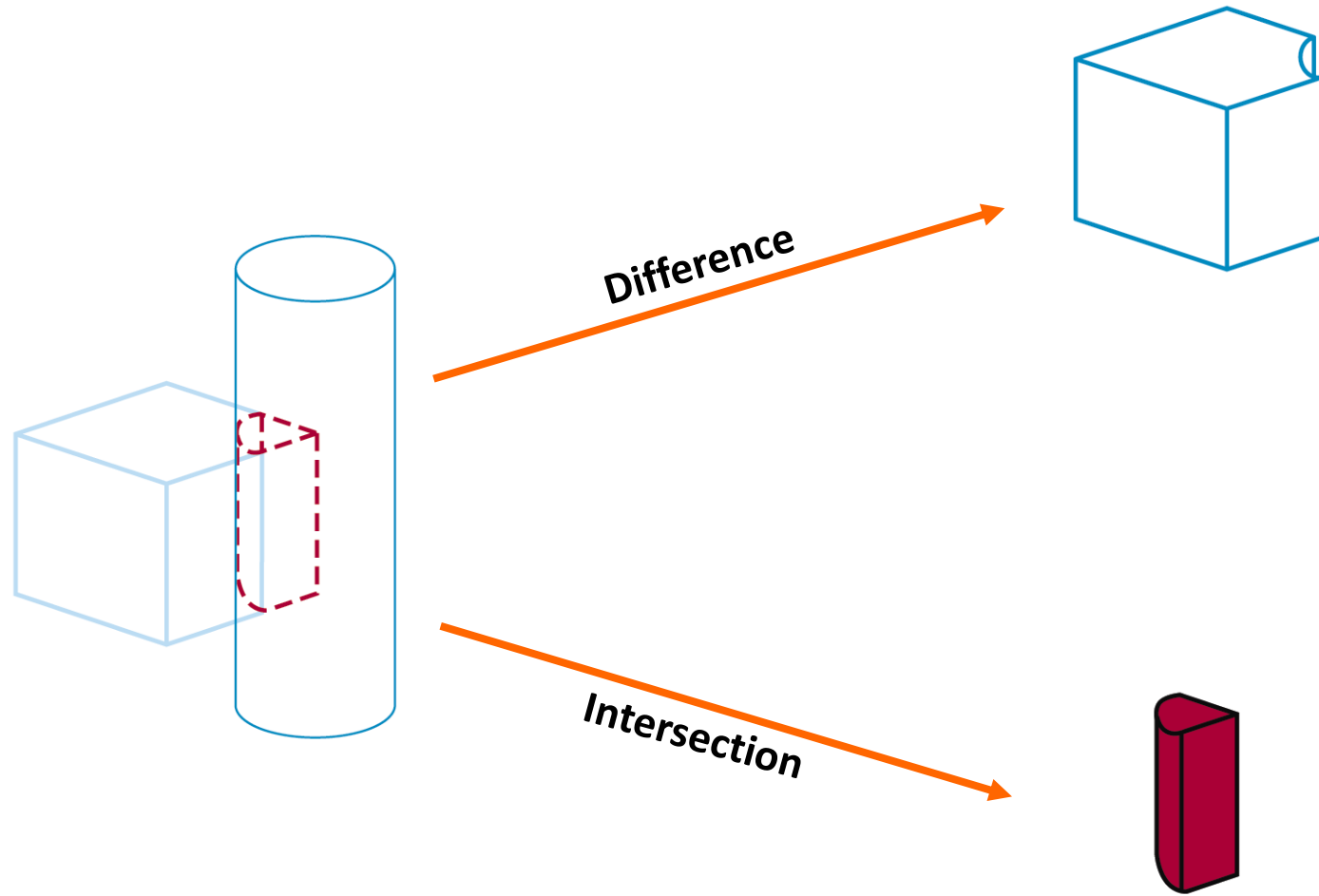


$A \cap B$



$B - A$

Constructive Solid Geometry Methods

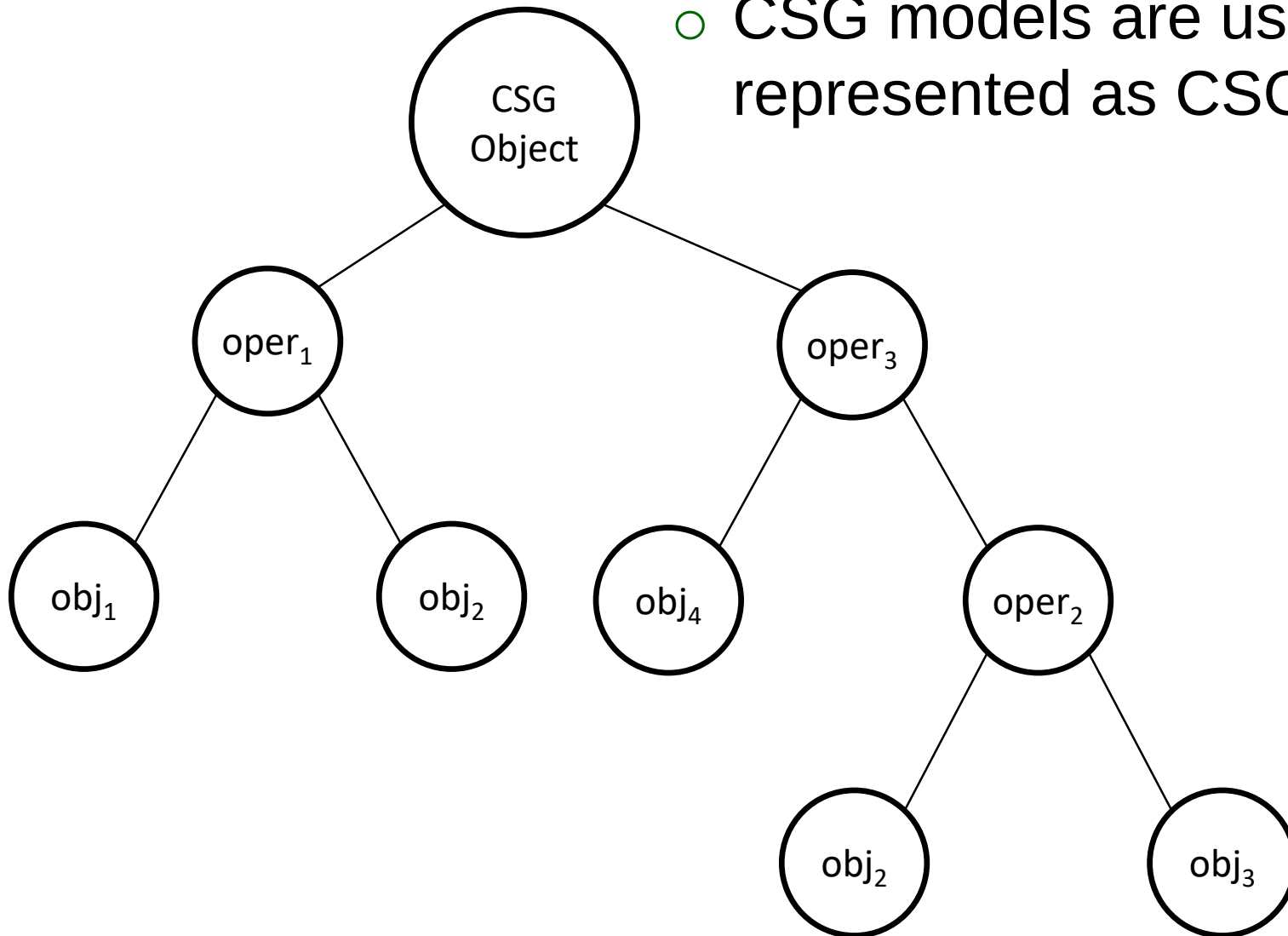


Constructive Solid Geometry Methods

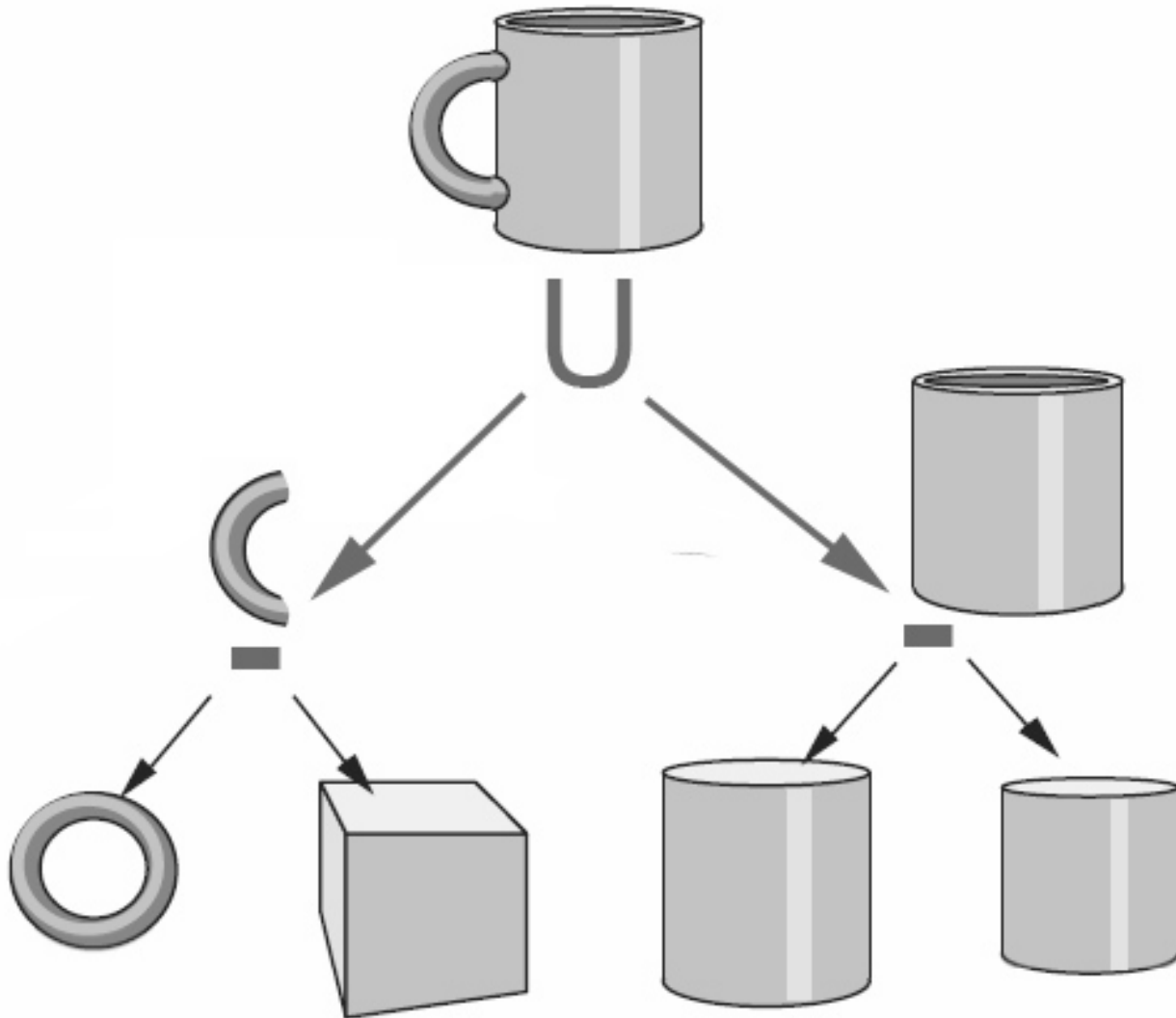
- CSG usually starts with a small set of primitives such as blocks, pyramids, spheres and cones
- Two objects are initially created and combined using some set operation to create a new object
- This object can then be combined with another primitive to make another new object
- This process continues until modeling complete

Constructive Solid Geometry Methods

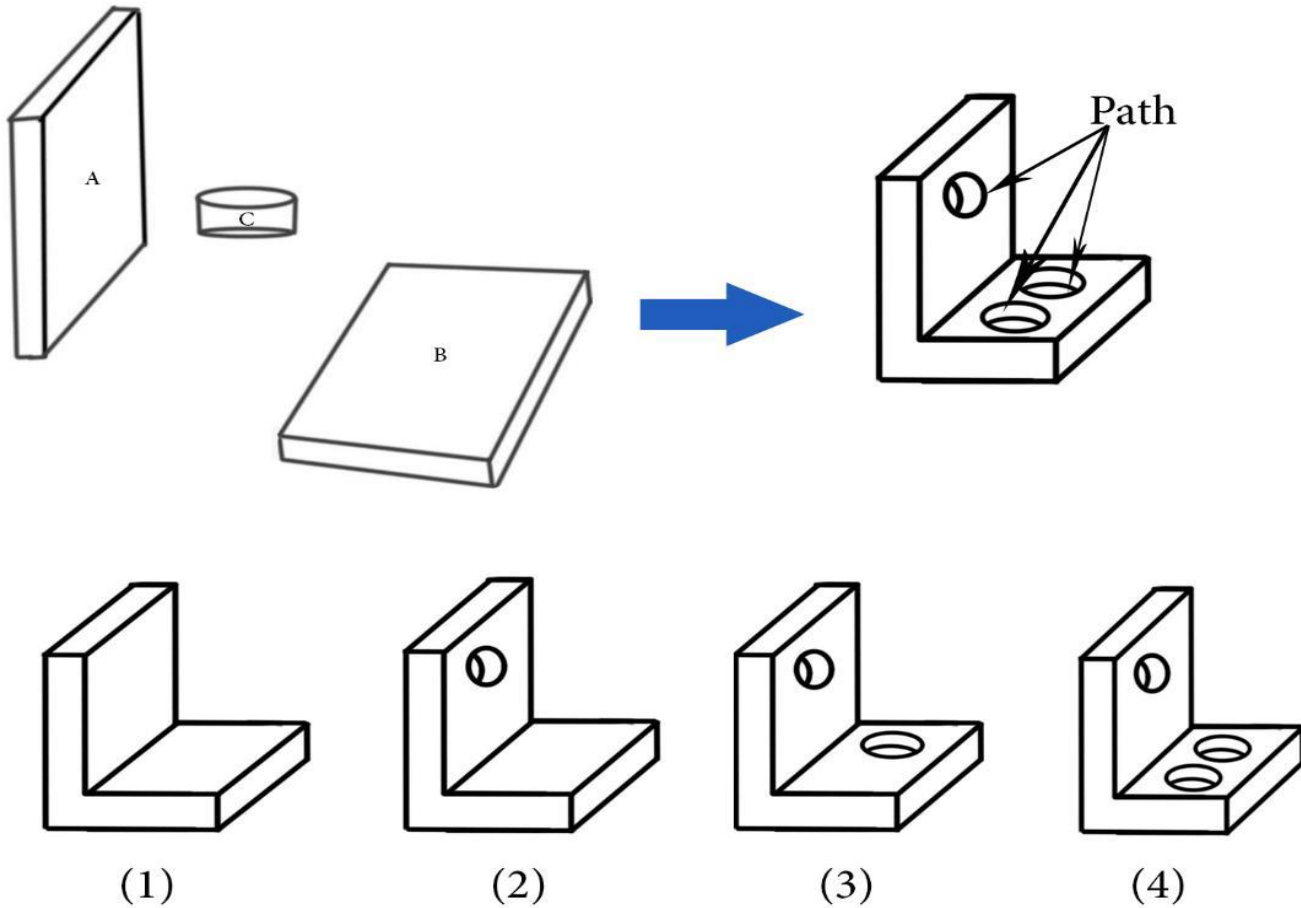
- CSG models are usually represented as CSG trees



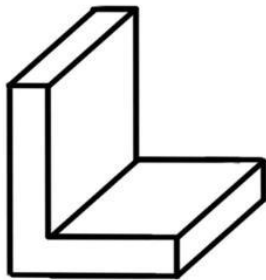
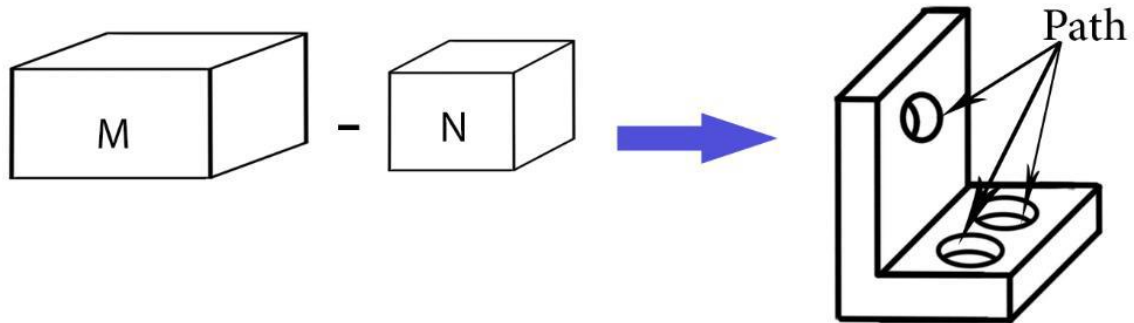
Constructive Solid Geometry Methods



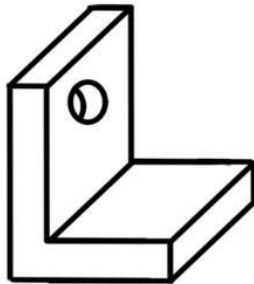
$$((((A \cup B) \setminus C) \setminus C) \setminus C)$$



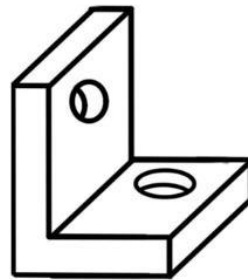
$$((((M \setminus N) \setminus C) \setminus C) \setminus C)$$



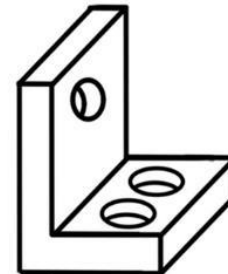
(1)



(2)



(3)



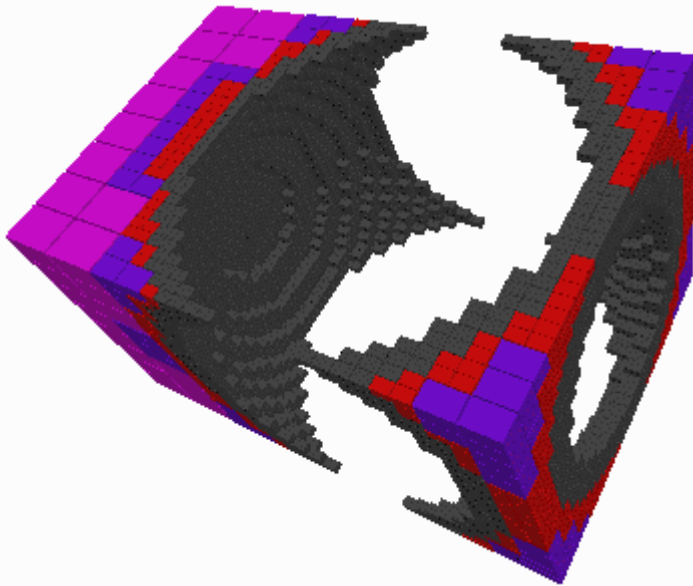
(4)

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Octrees

Octrees are hierarchical tree structures used to represent solid objects



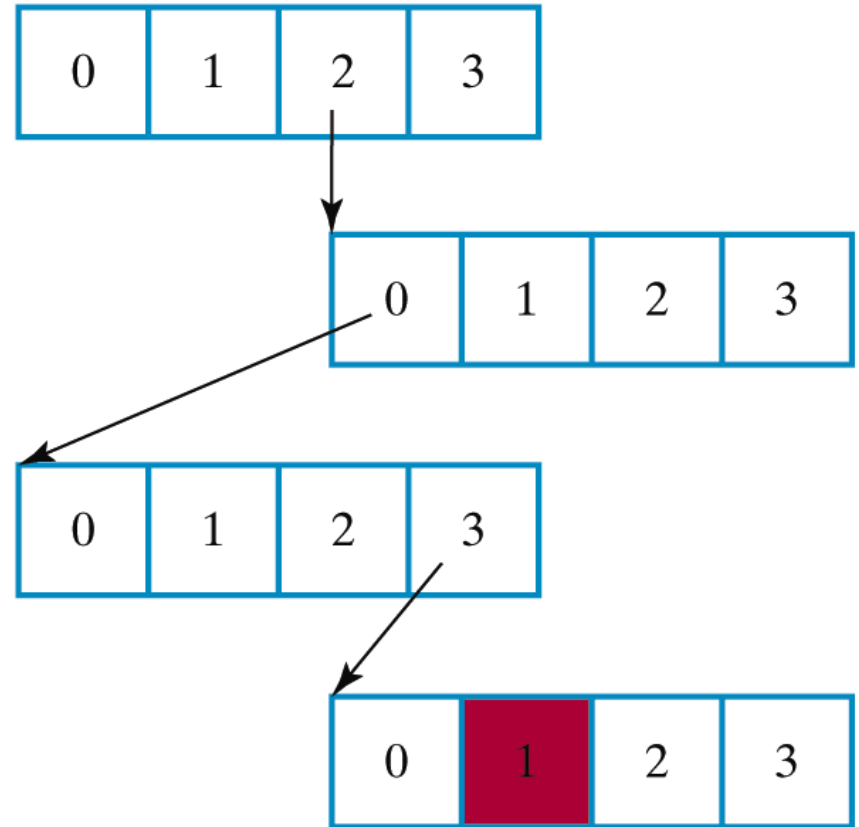
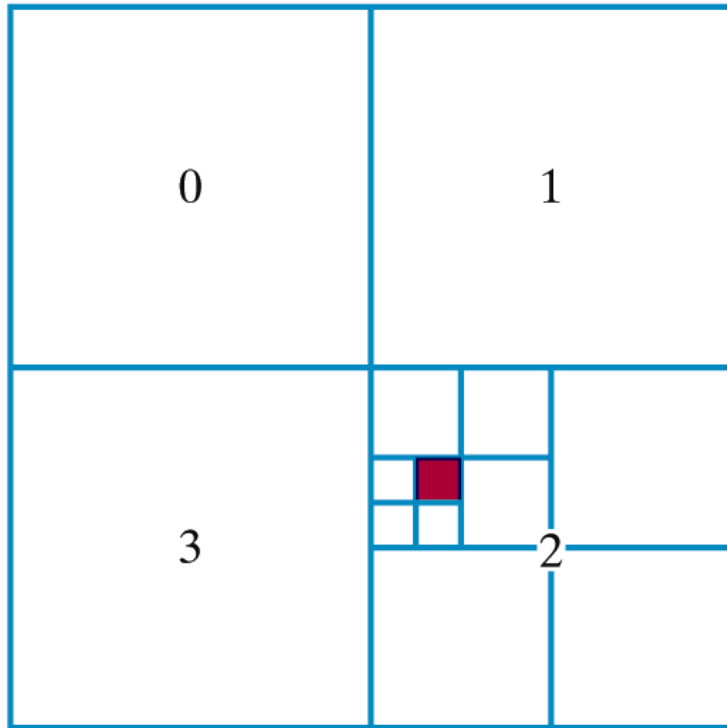
Octrees are particularly useful in applications that require cross sectional views – for example medical applications

Octrees are typically used when the interior of objects is important

Octrees & Quadtrees

- Octrees are based on a two-dimensional representation scheme called quadtree encoding
- Quadtree encoding divides a square region of space into four equal areas until homogeneous regions are found
- These regions can then be arranged in a tree

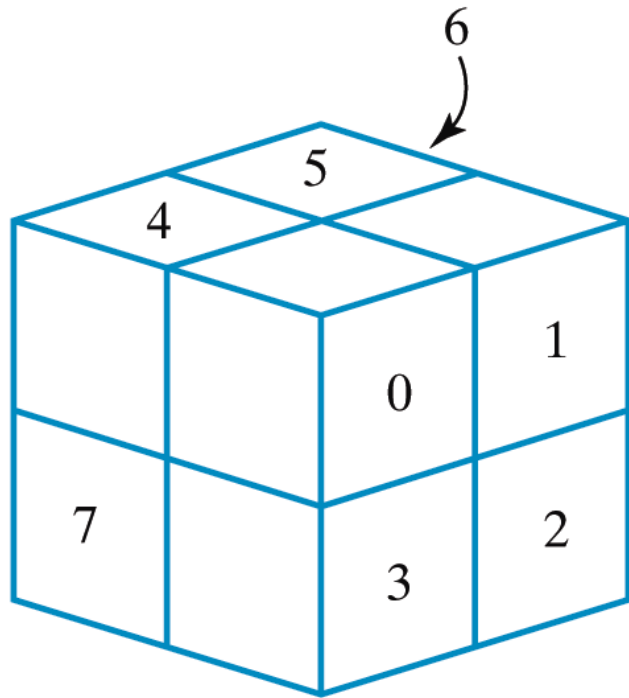
Quadtree Example



Octrees

- Quadtree encodings provide considerable savings in storage when large colour areas exist in a region of space
- An octree takes the same approach as quadtrees, but divides a cube region of 3D space into octants
- Each region within an octree is referred to as a volume element or voxel
- Division is continued until homogeneous regions are discovered

Octrees (cont...)



Region of a
Three-Dimensional
Space

0	1	2	3	4	5	6	7
---	---	---	---	---	---	---	---

Data Elements
in the Representative
Octree Node

Octrees (cont...)

- In 3 dimensions regions can be considered to be homogeneous in terms of colour, material type, density or any other physical characteristics
- Voxels also have the unique possibility of being empty

Octree Examples

